ENGINEERING PORTFOLIO

Stiven Kodra, Ph.D.

Mechanical Engineer
Additive Manufacturing · Process Development

st0kodra@gmail.com
linkedin.com/in/stiven-kodra

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02 M.S Research —Custom 3D Printer Design & Fabrication (ARME)

03 InkBit 3D — DfAM Optimization & Failure Analysis

04 Epoxy Curing Characterization — In Situ 3D-DIC Study

01 PhD RESEARCH

Compact Pellet Extruder for Sustainable Fused Granular Fabrication · UMass Lowell · 2022–2026

Article I

Compact Mixing Screw Design & Validation

Polymers 2025

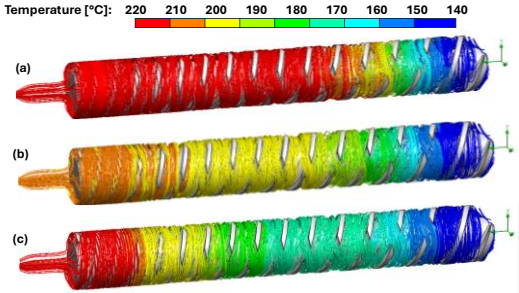


8:1 L/D · 7 prototype screws · SolidWorks Flow CFD · 0.74 kg/hr rPP · 56.5% thermodynamic efficiency

Article II

Steady-State Extruder Characterization

IJAMT 2026 (1st author)

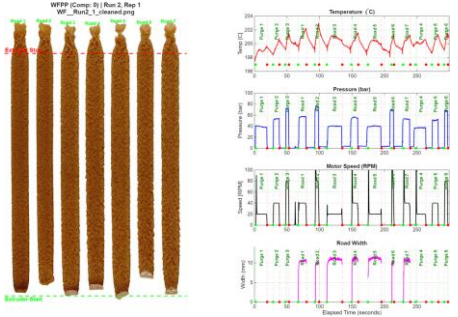


31 mm scale-up · 5.5 kg · 3 kg/hr · 76% efficiency · decoupled process gain matrix · orientation-agnostic

Article III

Transient Characterization & Feedforward Control

JMMP 2026 *(1st author)*



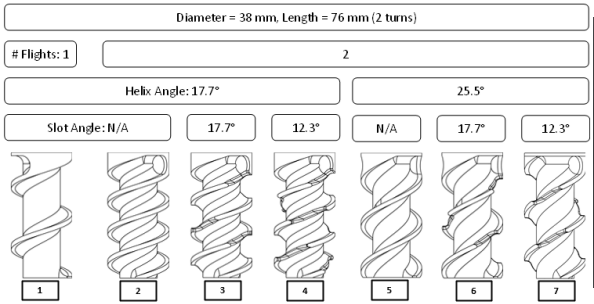
System ID: G(s)=0.990/(1+1.909s) · RPP1 & WFPP · G-code post-processor · latency IQR 5–20 mm → 2–8 mm

Funded by U.S. Army SBIR Phase II via re:3D Inc.

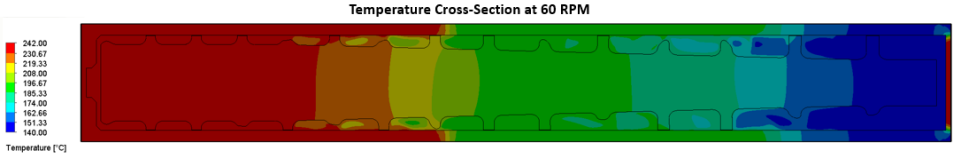
Compact Mixing Screw — Design & Simulation Validated at 8:1 L/D

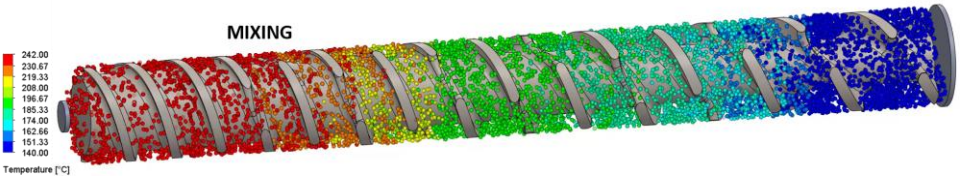
Article I · Polymers 2025 · Kazmer & Kodra

Skills: SolidWorks Flow Simulation · Screw design · MATLAB regression · Recycled polymer processing









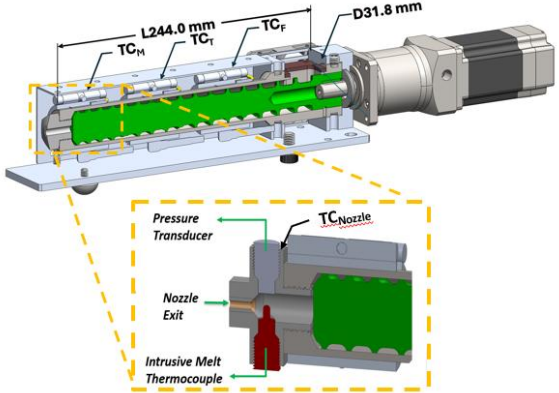
Particle Tracking and Mixing Visualization

- 7 prototype screw sections (SLS, PA12) — varied flight count, helix angle, mixing slot depth
- SolidWorks Flow Simulation: 432K–1M+ cell non-isothermal CFD validated at 14 & 34 RPM
- Final: double-flighted 20° helix, counter-rotating mixing slots — induces chaotic advection
- Pilot screw machined in 316 SS; instrumented for mass output & SEC measurement
- MATLAB regression: mass output $R^2 = 0.977$ · SEC $R^2 = 0.950$
- 0.74 kg/hr at 40 RPM (rPP) — effective conveying at just 8:1 L/D
- SEC: 0.264 kWh/kg HIPS · 0.344 kWh/kg rPP — outperforms 27:1 L/D barrier screws
- CFD temp within ~10 °C; mass flow within ~30% (packing density effect identified)

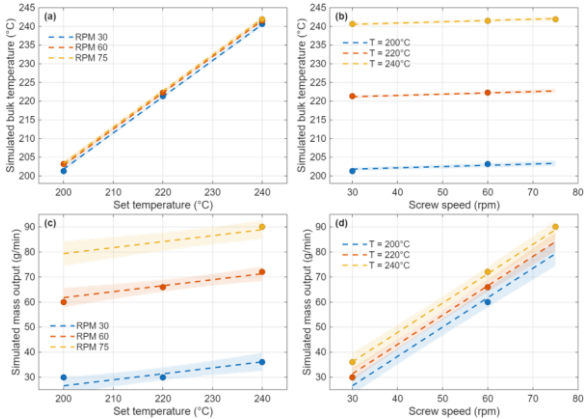
31 mm Scale-Up Achieves 3 kg/hr — Decoupled Process Gain Confirmed

Article II · IJAMT 2026 · Kodra & Kazmer · First Author

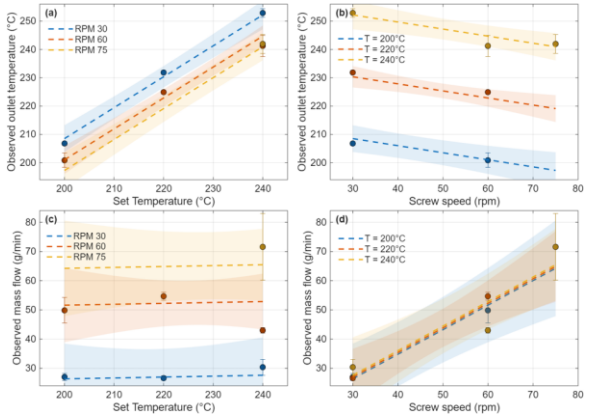
Skills: Full-factorial DOE · CFD · Instrumented nozzle design



- 5.5 kg System Mass — under 10 kg gantry limit
- 4 independent heating zones · 20:1 planetary gearbox
- 11-run DOE: screw speed (30–75 RPM), temp (200–240 °C), nozzle size
- SolidWorks Flow CFD (829K cells) validated against experimental data
- Process gain matrix via MATLAB fitlm — confirmed near-decoupled control
- 3 kg/hr at 60 RPM — design throughput target met
- Screw speed → mass flow: 0.84 g/min per RPM
- Barrel temp → melt temp: 1.1 °C per °C
- 76% thermodynamic efficiency



Simulation Gain Matrix

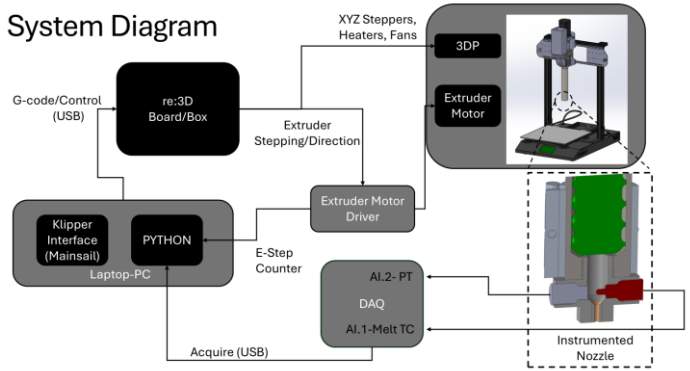


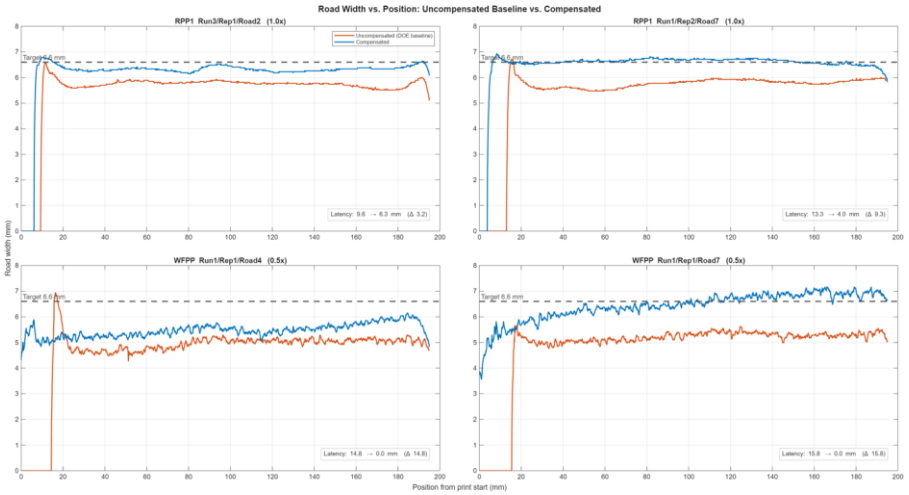
Experimental Gain Matrix

Feedforward Compensator Cuts Print Latency from 20 mm to 2 mm

Article III · JMMP 2026 · Kodra & Kazmer · First Author

Skills: System identification (MATLAB) · Python DAQ · Image processing · Feedforward G-code development





- V2 screw + custom gantry at 4.7 kg — installed on Creality CR-10S Pro
- Two PCR feedstocks:
  - rPP (RPP1)
  - 20 wt% wood-fiber PP composite (WFPP)
- Pressure Transducer + thermocouple at 100 Hz via 24-bit DAQ screw position at 100 Hz (Python)
- Road geometry measured by 600 DPI flatbed scans via MATLAB image pipeline
- System ID via MATLAB SIT (PEM): G(s) = 0.990 / (1 + 1.909s) · hardware-agnostic G-code compensator
- Material type dominant latency factor: F = 36.56, p < 0.001
- Compensator deployed as G-code post-processor — no firmware modification needed

Material	Run/Rep/Road	Method	Latency before	Latency after	Latency drop	Std of Width before	Std of Width after
RPP1	3/1/2	1.0x	9.55	6.30	3.25	0.21	0.19
RPP1	1/2/7	1.0x	13.30	4.00	9.30	0.25	0.24
WFPP	1/1/4	0.5x	14.75	0.00	14.75	0.34	0.26
WFPP	1/1/7	0.5x	15.80	0.00	15.80	0.22	0.56

02 MS Research — Custom 3D Printer - ARME (Additive Ram Material Extruder)

Additive Ram Material Extruder · Custom Machine Design & Fabrication · UMass Lowell · 2020–2023

ARME v1

Co-designed with Kazmer
Fabricated by Stiven

NEMA17 stepper
50:1 Gearbox
ACME ½×10 lead screw
10 kN load cell · 24 MPa max
Custom firmware integration

Polymer Composites 2021

Designed to allow for FDM additive manufacturing of highly viscous thermoset materials on desktop 3D printers

ARME 3XL (v2/v3)

Independent design
& fabrication

NEMA23 stepper
50:1 gearbox
Dual lead screws · ACME 3/8×16 lead screw
5 kN load cell · 65 MPa max
Dual-material
- Thermoplastic filament & Thermoset rubber
Ender 5 Pro firmware integration
Custom firmware integration

Polymers 2025 · Micromachines 2024

Improved designed to allow for multi-material processing of higher viscosity thermoset and tradition thermoplastic filament materials

ARME 4.0 (v4)

Independent design
& fabrication

NEMA23 stepper
50:1 gearbox
Dual lead screws · ACME 3/8×16 lead screw
Dual soft-material heads
2x 5 kN load cell · 65 MPa max
2 in^3 capacity per head
Dual-extrusion synchronization
Custom firmware integration

Currently used in University Research

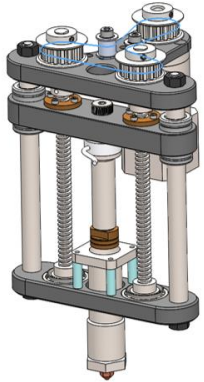
Redesigned to allow for dual soft material printing

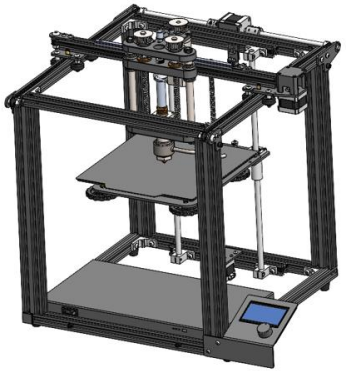
NSF SHAP3D I/UCRC • Grant #IIP-1822147 • 5 publications across lineage • Patent: U.S. App. 18/112,652

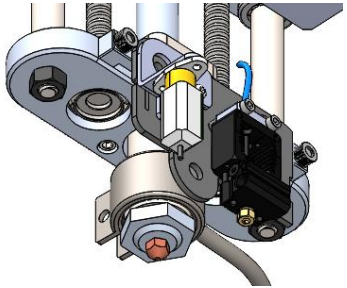
From 24 MPa to 650 Bar — Four Hardware Generations, Independently Designed

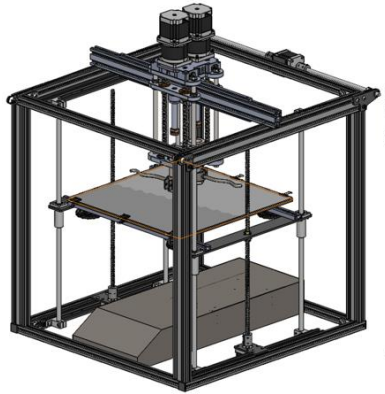
ARME v1 → v4 Hardware Detail · UMass Lowell

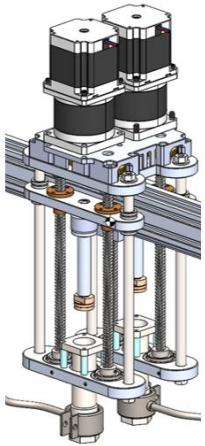
Skills: Piston extruder design · Custom firmware integration (Creality/Klipper) · Dual-material system design

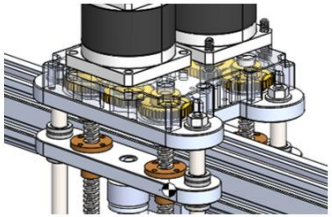












ARME V2 — CORE REDESIGN

Piston with 0.02 mm clearance + dual annular seal grooves

ACME 3/8 x16 lead screw + tapered roller bearing + geared NEMA17 · 1 step = 0.18 mm^3 output

Titanium standoffs and cored piston rod for thermal isolation

Firmware: reconfigured Creality E-axis as virtual 1.75 mm filament extruder

ARME 3XL & 4XL— INDEPENDENT REDESIGN

NEMA23 + 50:1 gearbox · dual symmetric lead screws — 3× barrel capacity (15 cm^3)

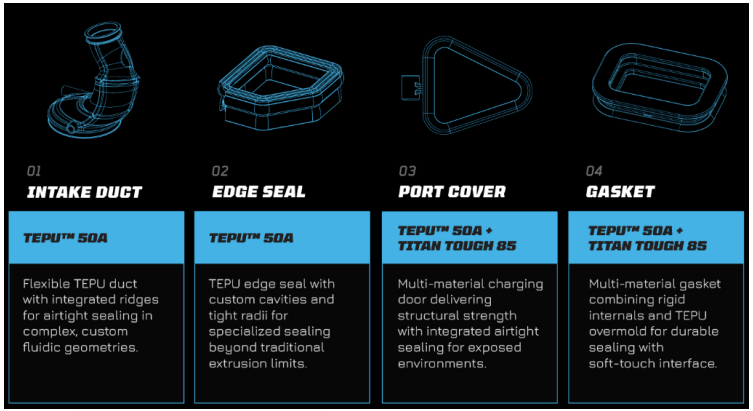
18.6 kN max force → 650 bar · Mosquito hot end for simultaneous rubber + FDM filament

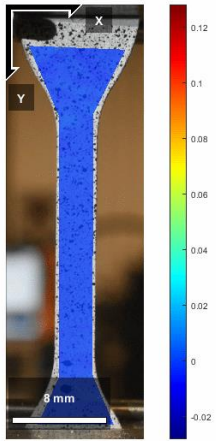
Full firmware integration — Ender 5 Pro · synchronized dual-material extrusion via G-code

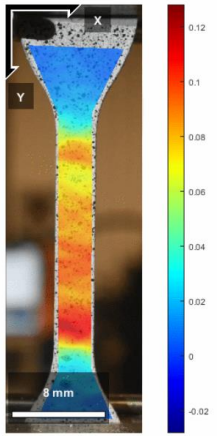
3.6 kg assembly mass · center of mass along linear slide bearing for gantry stability

03 INKBIT 3D — DfAM Optimization & Failure Analysis

Application Engineer · Medford, MA · 2022–Present · Multi-material inkjet printing







DFAM OPTIMIZATION & Demo Part Creation

Redesigned complex multi-material injection-molded components for inkjet AM production

Engineered custom mechanical interlocking geometries ensuring structural integrity between dissimilar materials

Optimized CAD models and Design Guides for factory throughput — supporting $200k+ in direct part sales

CAD and CAE tools: OnShape, proprietary Inkbit workflows

DIC FAILURE ANALYSIS

2D-DIC root-cause analysis on printed parts — de-correlated process defects from actual failure modes

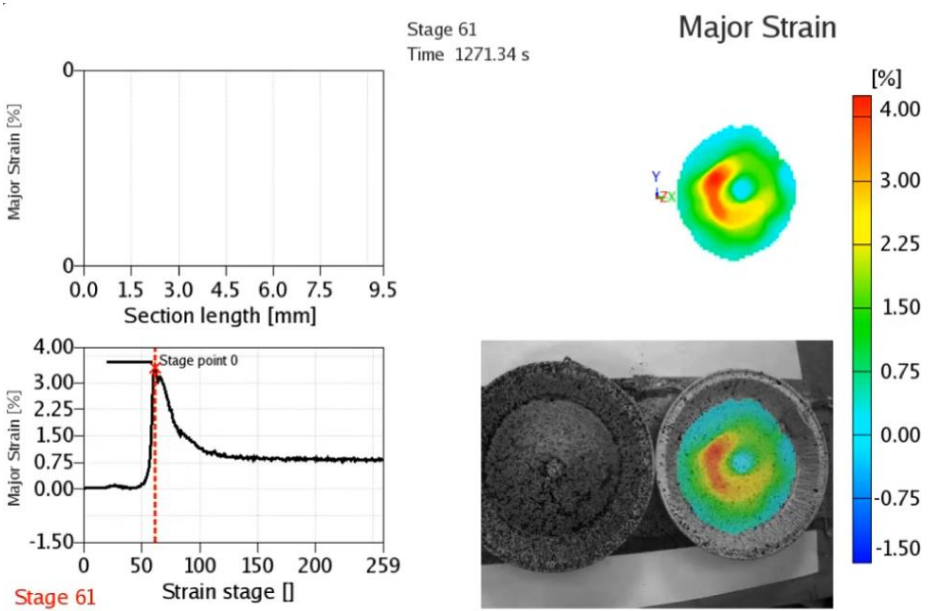
Prevented unnecessary process changes by identifying true failure initiation sites

Performed mechanical and material analysis to guide high-value client decisions

Skills: DfAM · Multi-material design · 2D-DIC failure analysis · CAD · CAE · Technical sales support

04 EPOXY CURING — In Situ 3D-DIC Characterization

Directed Study · Dr. Marianna Maiaru · UMass Lowell · Fall 2021 · Hexion EPON 862 / Epikure 3234



OBJECTIVE

Determine if 3D-DIC can capture the evolution of fiber/matrix interfacial strain in situ during epoxy cure — relevant to residual stress buildup in polymer matrix composites manufacturing

EXPERIMENTAL APPROACH

7 experiments across 3 container designs — iterated to eliminate boundary effects and speckle pattern failures

Carbon rod (10 mm Ø) embedded in Hexion EPON 862 / Epikure 3234 (100:14 ratio)

Baumer FWX201 stereovision DIC cameras (2 MP, 12 mm focal length) · images every 30 s for 2–3 hrs

K-type thermocouples logged resin internal temperature and ambient simultaneously

MATLAB pipeline: loaded DIC stage files, extracted max major strain per stage, animated vs. temperature

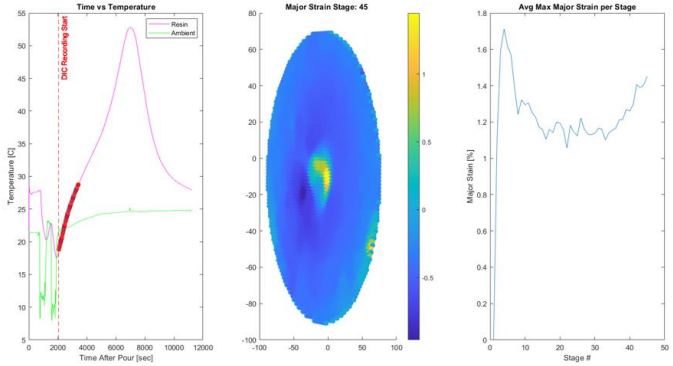
Skills: 3D-DIC · Epoxy resin processing · Thermocouple instrumentation · MATLAB data pipeline

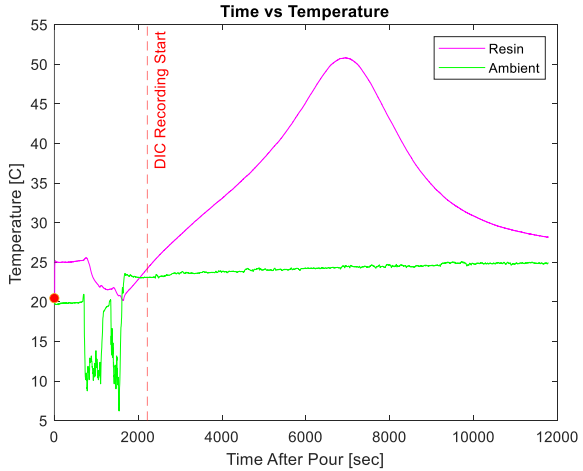
Strain Field Correlated with Cure Exotherm — Gelation Identified at Stage 45

Experiment 6 Results · Single Rod, 220 mm Pan

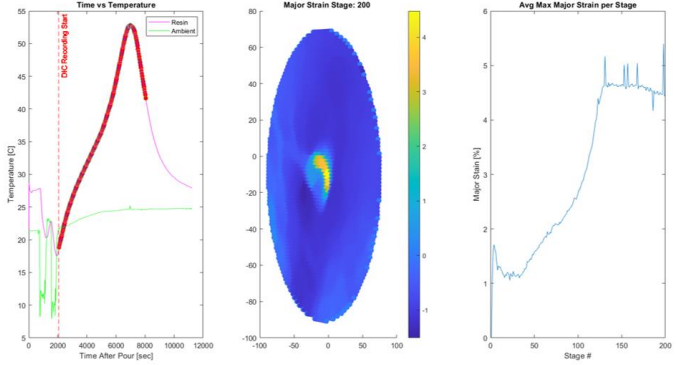
Finding: 3D-DIC can reliably capture in situ interfacial strain evolution during thermoset cure with proper setup

Exp. 6 Stage 45 (70 min)





Exp. 6 Stage 200 (final)



KEY FINDINGS

- Rapid strain increase at stage 45 (~70 min) — identified as gelation point of EPON 862
- Strain plateau and slight reduction after stage 130 (~98 min, 80% of peak exotherm)
- Major strain localized around carbon rod — confirms interfacial stress concentration during cure
- Experiment iteration produced clean, repeatable speckle patterns in final 220 mm pan setup
- MATLAB pipeline enabled animated strain visualization synced to live temperature data